1. Project Context

Imagine that you are on a team that has been asked to design a system that should be able to receive an electrical control signal and move the center of mass of the system from an origin point to a destination point and back again to the origin point on board the International Space Station (ISS). The below specifications and requirements have been created by your team.

Your task is to evaluate the requirements below using the provided ConOps and the requirements evaluation form. Please **form groups of two** and evaluate the requirements to the best of your ability using the material you have learned in class so far (see Lecture 14, Slides 26 and 31).

2. Concept of Operations

The system **must** be capable of performing two high-level operations:

1. Displace the center of mass of the system from its origin to a set destination.
2. Return that point mass back to its origin.

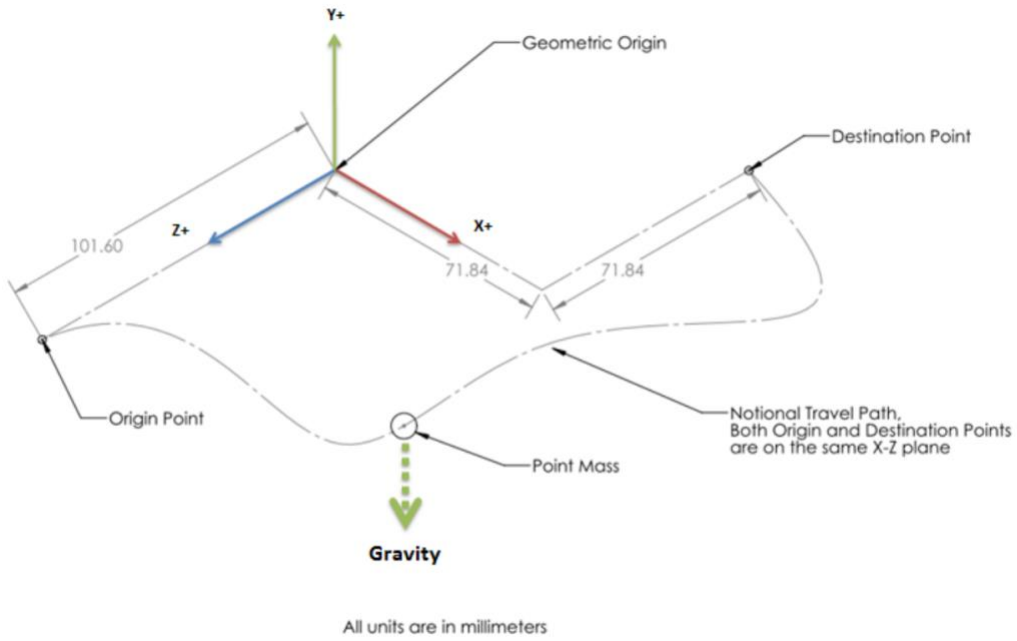


Figure 1: Concept of Operations

3. Requirements

3.1 Functional Requirements

FR1 Displace: The System shall be able to move a point mass from an origin point to a fixed destination point as seen in Figure 1. **Note 1:** The path shown in the figure is notional; the actual path is to be determined. **Note 2:** Assume that a link between the system output and the point mass will be designed separately (by someone else) and will match the needs of your design. **Note 3:** The origin point is (X, Y, Z) = (0 mm, 0 mm, 101.6 mm) or [0", 0", 4"]. **Note 4:** The destination point is (X, Y, Z) = (71.84 mm, 0 mm, -71.84 mm) or [2.82", 0", -2.82"].

FR2 Return: The System shall be able to move the point mass from the destination point back to the origin point.

3.2 Performance Requirements

PR1 The system shall execute FR1 accurately with +/- 2.5mm [0.1"] accuracy in every Cartesian direction.

PR2 The system shall execute FR2 accurately with +/- 2.5mm [0.1"] accuracy in every Cartesian direction.

PR3 The System shall be able to execute FR1 in no more than 30 seconds.

PR4 The System shall be able to execute FR1 in no less than 5 seconds.

PR5 The System shall be able to execute FR2 in no more than 30 seconds.

PR6 The System shall be able to execute FR2 in no less than 5 seconds.

PR7 The system shall be able to move a point mass of 3.6 kg perpendicularly to normal gravity (9.81 m/s^2).

3.3 Environmental Requirements

ER1 The System shall operate in an atmosphere comparable to that of Earth. Assume 21 degrees centigrade [70 degrees Fahrenheit], with low humidity, and pressurized to 100 kPA [750 mm Hg, 14.5 psi].

ER2 The System shall not contribute any particulates (e.g. dust) to the ISS atmosphere.

ER3 The System shall enclose all lubricated components to prevent lubricants from leaking into the atmosphere of the ISS.

3.4 Resource Requirements

RR1 The System shall not use more than 1.5 Watt-hours to support all operations. **Note:** Assume a maximum of 1 minute of operations.